1 Collective Communication

The simplest collective communication function is `MPI_Barrier`, which blocks until all processes in the communicator have reached the barrier.

```
int MPI_Barrier(MPI_Comm comm)
```

The broadcast function `MPI_Bcast` sends data from one process to all others.

```
int MPI_Bcast(void* buffer, int count, MPI_Datatype datatype,
int root, MPI_Comm comm)
```

2 More Collective Operations

The gather function `MPI_Gather` collects data from all processes.

```
int MPI_Gather(const void* sendbuf, int sendcount, MPI_Datatype sendtype,
void* recvbuf, int recvcount, MPI_Datatype recvtype,
int root, MPI_Comm comm)
```

The scatter function distributes data from one process to all others.

```
int MPI_Scatter(const void* sendbuf, int sendcount, MPI_Datatype sendtype,
void* recvbuf, int recvcount, MPI_Datatype recvtype,
int root, MPI_Comm comm)
```